MVP HARDWARE BUILD GUIDE

Precision Agriculture Monitoring Quadruped Robot Dog

System Architecture • 3D Print Files • Bill of Materials
Assembly Guide • Wiring • Power Distribution • Software Setup

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Programme: B.Tech — CSE (AI & Robotics)

Academic Year: 2025 – 2026

Version: MVP v1.0 — Compact Quadruped Prototype

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1. Project Overview & MVP Design Philosophy

This document is a complete hardware build guide for constructing a Minimum Viable Product (MVP) version of the Precision Agriculture Monitoring Quadruped Robot Dog. The MVP prioritizes functional demonstration over perfection — the goal is a small, compact, fully working quadruped that can walk, carry all sensors, run AI inference on the Jetson Nano Orin, and transmit telemetry data.

1.1 MVP Design Principles

- **Compact Size:** Keep the robot small (body ≈ 20 × 14 × 8 cm) to minimize 3D print time and material cost.
- **Cheap Materials:** Use PLA+ for structural parts (strong, cheap, easy to print). Use TPU for foot pads (flexible grip).
- **Commodity Servos:** Use readily available MG996R servo motors (~₹250 each). Strong enough for a small robot, dramatically cheaper than BLDC motors.
- **Jetson-First Design:** Body sized precisely to mount the Jetson Nano Orin Developer Kit (69.6 × 45 mm), PCA9685 driver board, and a compact LiPo battery.
- **Sensor Integration:** Sensors mount on the front/top of the body — camera forward-facing, environmental sensors exposed to airflow.
- **Printable on Budget Printers:** Every part is designed to print without supports on a standard 220 × 220 mm print bed (Ender 3 / Prusa Mini compatible).

1.2 Target Specifications (MVP)

Parameter	MVP Target
Overall Body Dimensions	~200 × 140 × 80 mm (L × W × H)
Standing Height (to body)	~150–180 mm
Leg Span (hip to foot)	~120 mm per leg
Total Weight (with electronics)	~1.2 – 1.5 kg
Degrees of Freedom (DOF)	12 (3 per leg × 4 legs)
Servo Type	MG996R (or DS3218 for extra torque)
Compute Board	NVIDIA Jetson Nano Orin Developer Kit
Battery	11.1V 3S LiPo 2200–3000 mAh
Runtime Target	~15–20 minutes walking
3D Print Material	PLA+ (body) / TPU (feet)
Estimated Total Cost	₹35,000 – ₹45,000

2. System Architecture (Detailed)

The MVP system architecture consists of four interconnected tiers. Each tier is physically modular, meaning components can be individually removed, debugged, or upgraded without disassembling the entire robot.

2.1 Perception Tier (Sensors)

All sensors interface with the Jetson Nano Orin via standard protocols (I^2C, UART, CSI, GPIO). Sensors are mounted on the front face and top deck of the body chassis.

Sensor	Model	Interface	Measurement	Mounting Position
Camera	IMX219 (8MP CSI)	MIPI CSI-2	RGB Images 3280×2464	Front face, angled 15° down
Dust/PM2.5	PMS5003	UART (Serial)	PM1.0, PM2.5, PM10 ($\mu g/m^3$)	Top deck, exposed to airflow
Gas (NH_3/VOC)	MQ-135	Analog (via ADS1115 ADC)	NH_3, Benzene, CO_2 (ppm)	Top deck, open air
Temp & Humidity	DHT22 (AM2302)	Digital GPIO (1-wire)	Temp (°C), Humidity (%RH)	Top deck, shaded slot
IMU (Balance)	MPU6050	I^2C (0x68)	Accel (3-axis), Gyro (3-axis)	Center body, flat mounted
GPS	NEO-6M (u-blox)	UART (Serial)	Lat, Lon, Altitude, Speed	Top deck, antenna skyward
Ultrasonic	HC-SR04	GPIO (Trig/Echo)	Distance (2–400 cm)	Front face, beside camera

2.2 Compute Tier (Jetson Nano Orin)

The NVIDIA Jetson Nano Orin is the central brain of the robot. It runs all AI inference, sensor data aggregation, inverse kinematics calculations, and communication protocols simultaneously.

- **Board:** NVIDIA Jetson Orin Nano Developer Kit (8GB)
- **CPU:** 6-core ARM Cortex-A78AE CPU @ 1.5 GHz
- **GPU:** 1024-core NVIDIA Ampere GPU (up to 40 TOPS AI performance)
- **RAM:** 8 GB 128-bit LPDDR5 (shared CPU/GPU)
- **Storage:** MicroSD (64 GB Class 10 A2 recommended) or NVMe M.2 SSD
- **OS:** Ubuntu 20.04 LTS with JetPack 6.x SDK
- **Middleware:** ROS 2 Humble Hawksbill

Compute Task Allocation:

Task	Runs On	Target Frequency
Gait Controller & IK	CPU (Cortex-A78AE)	50 Hz (20 ms loop)
Disease Detection CNN	GPU (Ampere) via TensorRT	30+ FPS
Sensor Data Aggregation	CPU	10 Hz
IMU Balance PID Controller	CPU	100 Hz
MQTT Telemetry Publisher	CPU	2 Hz
Camera Frame Capture	ISP → GPU	30 FPS
Path Planning (Nav2)	CPU	On-demand

2.3 Actuation & Power Tier

The actuation system converts computed joint angles from the IK engine into physical servo positions. Power management ensures stable, isolated voltage rails for logic and motor systems.

- **Servos:** 12 × MG996R (or DS3218 for higher torque) — 3 per leg × 4 legs.
- **Servo Driver:** PCA9685 16-Channel PWM Driver Board (I^2C address 0x40).
- **Battery:** 11.1V 3S LiPo, 2200–3000 mAh, 25C discharge rate.
- **Power Regulation:** Dual buck converters: one 6V/5A rail for servos, one 5V/3A rail for Jetson + sensors.

2.4 Base Station / Dashboard Tier

The base station is a laptop or PC running a web dashboard. The robot communicates over Wi-Fi, publishing sensor data and disease detection alerts via MQTT protocol.

- **Communication:** MQTT Broker (Mosquitto) running on the Jetson or the base station PC.
- **Dashboard:** Flask (Python backend) + React or plain HTML/JS frontend.
- **Features:** Live camera feed (via MJPEG or WebRTC), environmental heatmaps, GPS field map, disease alert log.

3. MVP Dimensions & Form Factor

The MVP body is designed as a flat rectangular platform with four leg mounts at the corners. This maximizes internal volume for electronics while keeping the footprint compact. The robot uses a "mammalian" leg configuration (all knees bending the same way) for simplicity.

3.1 Key Dimensions

Dimension	Measurement	Notes
Body Length	200 mm	Fits Jetson + PCA9685 side-by-side
Body Width	140 mm	Clearance for servo mounts at corners
Body Height (shell)	80 mm	Two-piece clamshell (top plate + base plate)
Upper Leg Length	55–60 mm	Hip joint to knee joint
Lower Leg Length	60–65 mm	Knee joint to foot pad
Standing Height	150–180 mm	Ground to top of body (adjustable via IK)
Foot-to-Foot Span (lateral)	~280 mm	Provides stable lateral base
Foot-to-Foot Span (longitudinal)	~250 mm	Front feet to rear feet distance
Hip Offset (shoulder width)	25–30 mm	Servo horn to body edge

3.2 Leg Configuration (3-DOF per Leg)

Each leg has three joints controlled by individual servo motors:

1. **Joint 1:** Hip Abduction/Adduction — Swings the leg outward/inward (side-to-side). Servo mounted inside the body, horn protruding through side wall.
2. **Joint 2:** Hip Flexion/Extension — Swings the upper leg forward/backward. Servo mounted at the hip bracket.
3. **Joint 3:** Knee Flexion/Extension — Bends the lower leg. Servo mounted at the knee bracket.

4. 3D Printed Parts — Complete File List

Below is the complete list of 3D printable STL files required to build the MVP quadruped. These can be designed in Fusion 360, SolidWorks, or FreeCAD. Open-source designs from projects like SpotMicro, OpenQuadruped, or MangDang's Mini Pupper can serve as starting references.

4.1 Body / Chassis Parts

#	Part Name (STL File)	Qty	Material	Description
1	body_base_plate.stl	1	PLA+	Bottom plate of chassis. Holds battery tray and servo wire channels. Includes 4 corner leg mount bosses.
2	body_top_plate.stl	1	PLA+	Top cover/shell. Snaps or screws onto base plate. Mounting holes for Jetson Nano Orin, PCA9685, and sensor breakout headers.
3	body_front_plate.stl	1	PLA+	Front face panel. Camera mount slot (IMX219) and ultrasonic sensor (HC-SR04) cutouts.
4	body_rear_plate.stl	1	PLA+	Rear face panel. Power switch cutout, XT60 charging port access, and ventilation grille.
5	battery_tray.stl	1	PLA+	Sliding tray for 3S LiPo battery. Velcro strap slots. Fits under the base plate.
6	sensor_top_mount.stl	1	PLA+	Top deck sensor platform. Mounting slots for DHT22, PMS5003, MQ-135, and GPS antenna.
7	jetson_standoffs.stl	4	PLA+	M2.5 standoff pillars to mount Jetson Nano Orin to the top

				plate with vibration isolation.
8	pca9685_bracket.stl	1	PLA+	Bracket to mount PCA9685 servo driver board beside the Jetson on the top plate.

4.2 Leg Parts (×4 sets — print 4 of each)

#	Part Name (STL File)	Qty	Material	Description
9	hip_bracket.stl	4	PLA+	L-shaped bracket connecting the hip ab/ad servo (body) to the hip flex/ext servo. Includes servo horn mount and M3 screw holes.
10	upper_leg.stl	4	PLA+	Upper leg link. Connects hip flex/ext servo horn to the knee servo. Length: 55–60 mm. Lightening slots for weight reduction.
11	lower_leg.stl	4	PLA+	Lower leg link. Connects knee servo horn to the foot. Length: 60–65 mm. Tapered profile for ground clearance.
12	foot_pad.stl	4	TPU 95A	Soft, flexible foot pad for ground grip and shock absorption. Hemispherical profile, 18–20 mm diameter. Printed in flexible TPU.
13	servo_horn_adapter.stl	12	PLA+	Custom adapter to rigidly couple the MG996R metal servo horn to the 3D printed leg links. Press-fit + M2 screw.

14	knee_cover.stl	4	PLA+	Protective cover for the knee servo. Prevents debris/soil ingress into the servo shaft.

4.3 Miscellaneous / Optional Parts

#	Part Name (STL File)	Qty	Material	Description
15	camera_mount_gimbal.stl	1	PLA+	Tilt mount for the IMX219 camera. Allows manual angle adjustment (0°–30° downward tilt).
16	wire_clip.stl	8	PLA+	Small press-fit cable clips to route servo wires neatly along the legs and into the body.
17	power_switch_holder.stl	1	PLA+	Recessed mount for the main ON/OFF rocker switch on the rear panel.
18	logo_name_plate.stl	1	PLA+	Optional decorative nameplate with project title for presentation display.

4.4 Total Print Summary

- **Total Unique Parts:** 18 unique STL files.
- **Total Pieces to Print:** ~46 individual printed components (including ×4 leg sets).
- **Estimated Total Filament:** ~350–450 grams PLA+, ~20 grams TPU.
- **Estimated Total Print Time:** ~18–24 hours on a standard FDM printer (Ender 3 / Prusa Mini).

5. 3D Printing Settings & Material Guide

5.1 Recommended Printer Specifications

- **Printer Type:** FDM (Fused Deposition Modeling) — Ender 3 V2, Creality K1, Prusa Mini+, or any 220×220 mm bed printer.
- **Nozzle:** 0.4 mm (standard).
- **Print Volume:** ≥ 220 × 220 × 250 mm (all parts fit without splitting).

5.2 Material Settings

Parameter	PLA+ (Body/Legs)	TPU 95A (Feet)	PETG (Optional)	ABS (Optional)	Notes
Nozzle Temp	210–220 °C	220–235 °C	230–250 °C	240–260 °C	PLA+ is easiest to print
Bed Temp	55–65 °C	50–60 °C	70–80 °C	90–110 °C	Use glue stick for TPU
Layer Height	0.2 mm	0.2 mm	0.2 mm	0.2 mm	0.16 mm for fine detail parts
Infill %	30–40%	50–60%	30–40%	30–40%	Higher for hip bracket & legs
Infill Pattern	Gyroid or Grid	Grid	Gyroid	Gyroid	Gyroid = best strength/weight
Wall Count	3–4 perimeters	3	3–4	3–4	4 walls for structural parts
Print Speed	50–60 mm/s	20–30 mm/s	40–50 mm/s	40–50 mm/s	Slow TPU to avoid jams
Supports	Minimal/None	None	Minimal	Tree Supports	Parts designed support-free

5.3 Color Recommendations (for Presentation)

- **Body Chassis:** Matte Black or Dark Grey — professional, hides layer lines.
- **Leg Links:** White or Light Grey — visual contrast with body.
- **Foot Pads (TPU):** Orange or Green — agricultural/robotic theme.

6. Bill of Materials (BOM) — All Components

The following table lists every component needed to build the complete MVP quadruped, including electronics, fasteners, and consumables. All prices are approximate Indian market rates (2025).

6.1 Compute & Electronics

Component	Model / Spec	Qty	Est. Price (₹)	Source
Jetson Nano Orin Dev Kit	NVIDIA Orin Nano 8GB	1	₹18,000–22,000	NVIDIA / Robu.in / Amazon
MicroSD Card	64 GB Class 10 A2	1	₹600	Amazon / Local
PCA9685 Servo Driver	16-Ch I^2C PWM Driver	1	₹200	Robu.in / Amazon
ADS1115 ADC Module	16-bit 4-Ch I^2C ADC	1	₹250	Robu.in (for MQ-135 analog)
Buck Converter (5V)	LM2596 5V/3A	1	₹80	Robu.in / Amazon
Buck Converter (6V)	LM2596 Adjustable (set to 6V)	1	₹80	Robu.in / Amazon
Power Switch	SPST Rocker Switch 10A	1	₹30	Local electronics shop
XT60 Connector Pair	Male + Female XT60	1 pair	₹50	Amazon / Hobby shop
Jumper Wires (M-F, F-F)	Dupont 20 cm assorted	3 packs	₹150	Robu.in
USB Wi-Fi Adapter	TP-Link / Realtek (if needed)	1	₹400	Amazon (only if built-in Wi-Fi is absent)

6.2 Sensors

Sensor	Model	Qty	Est. Price (₹)	Source
Camera	IMX219 8MP CSI Module	1	₹800–1,200	Robu.in / Amazon
Dust Sensor	PMS5003 Laser PM Sensor	1	₹1,200–1,500	Robu.in / AliExpress
Gas Sensor	MQ-135 Module	1	₹150	Robu.in / Amazon
Temp & Humidity	DHT22 (AM2302)	1	₹200	Robu.in / Amazon

	Module			
IMU	MPU6050 (GY-521 breakout)	1	₹150	Robu.in / Amazon
GPS Module	NEO-6M with antenna	1	₹400–500	Robu.in / Amazon
Ultrasonic Sensor	HC-SR04	1	₹50	Robu.in / Amazon

6.3 Actuators & Power

Component	Model / Spec	Qty	Est. Price (₹)	Source
Servo Motors	MG996R Metal Gear (13 kg·cm)	12	₹3,000 (₹250 each)	Robu.in / Amazon
Servo Motors (Alt)	DS3218 (20 kg·cm) — if more torque needed	12	₹6,000 (₹500 each)	Amazon / AliExpress
LiPo Battery	11.1V 3S 2200mAh 25C	1	₹1,200–1,500	Hobby King / Robu.in
LiPo Battery (Alt)	11.1V 3S 3000mAh 25C (longer runtime)	1	₹1,800–2,200	Hobby King / Amazon
LiPo Charger	B3 Compact Balance Charger	1	₹350	Amazon / Hobby shop
LiPo Voltage Monitor	1S-8S Battery Buzzer Alarm	1	₹80	Amazon (prevents over-discharge)

6.4 Fasteners & Hardware

Item	Specification	Qty	Est. Price (₹)
M2 Bolts + Nuts	6 mm, 8 mm, 10 mm assorted	~50 pcs	₹100
M2.5 Bolts + Nuts	8 mm, 10 mm (for Jetson mount)	~20 pcs	₹60
M3 Bolts + Nuts	8 mm, 10 mm, 12 mm (servo + legs)	~60 pcs	₹120
M3 Brass Inserts (heat-set)	4.5 mm OD × 5 mm length	~30 pcs	₹200
Servo Screws	Included with MG996R servo	—	Included

Velcro Straps	20 mm × 30 cm (battery strap)	2	₹50
Zip Ties	100 mm and 150 mm	1 pack	₹50
Silicone Wire	18 AWG Red + Black (power)	1 m each	₹100
Heat Shrink Tubing	Assorted 2–6 mm	1 set	₹80
Double-Sided Tape	3M VHB or foam tape	1 roll	₹100

6.5 3D Printing Filament

Material	Weight Needed	Color	Est. Price (₹)
PLA+ 1.75 mm	~400 g (from a 1 kg spool)	Black / Grey	₹800 (1 kg spool)
TPU 95A 1.75 mm	~25 g (from a 500 g spool)	Orange / Green	₹600 (500 g spool)

7. Servo Motor Selection & Leg Kinematics

7.1 Servo Comparison

Parameter	MG996R	DS3218	SG90 (Too Weak)	MG90S (Too Weak)	Recommendation
Torque	13 kg·cm @ 6V	20 kg·cm @ 6.8V	1.8 kg·cm	2.2 kg·cm	MG996R minimum; DS3218 if budget allows
Speed	0.13 sec/60°	0.16 sec/60°	0.12 sec/60°	0.08 sec/60°	All acceptable for crawl gait
Weight	55 g	60 g	9 g	13.4 g	MG996R/DS3218 acceptable for MVP
Gear Type	Metal Gear	Metal Gear	Plastic Gear	Metal Gear	Metal gears essential for longevity
Voltage	4.8–7.2 V	4.8–6.8 V	4.8 V	4.8–6 V	Use 6V regulated rail
Price (India)	~₹250	~₹500	~₹50	~₹120	MG996R for budget; DS3218 for performance
Stall Current	~2.5 A	~3.0 A	~0.7 A	~0.6 A	Size power supply accordingly

7.2 Inverse Kinematics (3-DOF Leg)

Each leg is modeled as a 3-DOF serial chain with the following joints:

- **Joint 1 (Shoulder):** θ_1 (hip abduction) — swings leg laterally. Range: ±30°.
- **Joint 2 (Hip):** θ_2 (hip flexion) — swings upper leg fore/aft. Range: ±60°.
- **Joint 3 (Knee):** θ_3 (knee flexion) — bends lower leg. Range: 0° to 120°.

Given a desired foot position (x, y, z) relative to the shoulder, the analytical IK solution uses:

- $\theta_1 = \text{atan2}(y, z)$ — for hip abduction angle.
- $D = \sqrt{(y^2 + z^2)}$ — d, where d is the hip offset length.
- $\theta_3 = \text{acos}((D^2 + x^2 - L_1^2 - L_2^2) / (2 \times L_1 \times L_2))$ — Law of Cosines for knee angle.
- $\theta_2 = \text{atan2}(x, D) - \text{atan2}(L_2 \times \sin(\theta_3), L_1 + L_2 \times \cos(\theta_3))$ — Hip flexion.

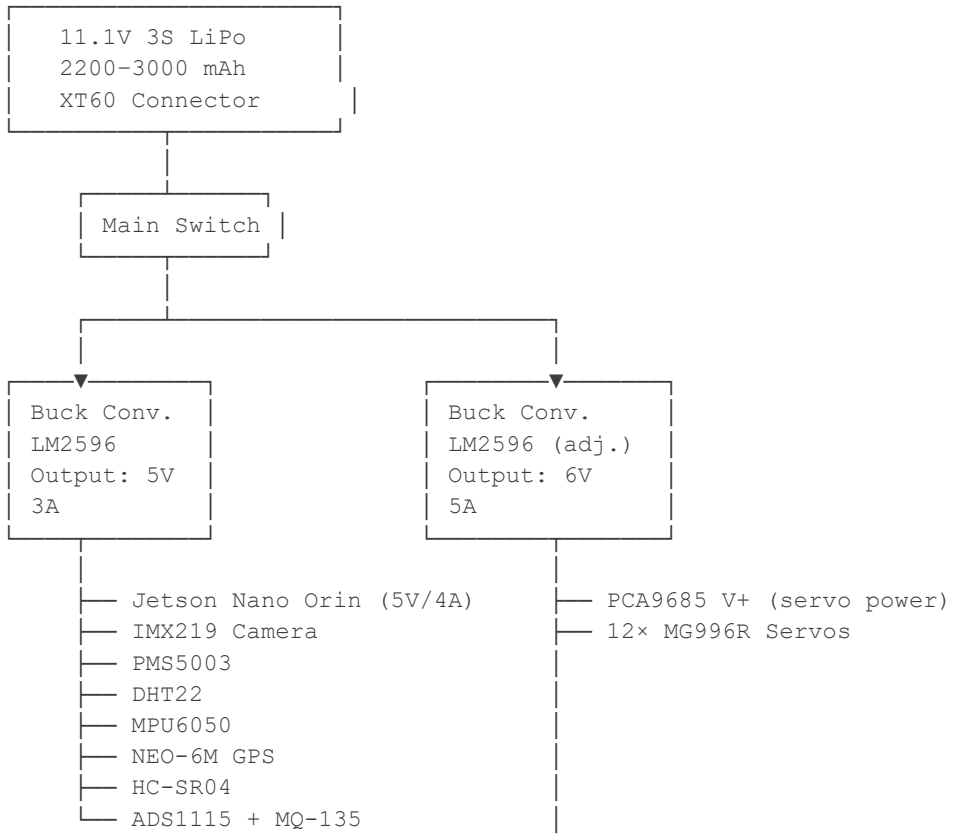
Where L_1 = upper leg length (55–60 mm), L_2 = lower leg length (60–65 mm). This closed-form solution computes in <0.1 ms per leg, allowing 50+ Hz gait loops on the Jetson CPU.

8. Power Distribution & Battery System

8.1 Power Architecture

The battery feeds two isolated buck converter rails. This prevents servo motor back-EMF spikes and high-current transients from causing brown-outs or reboots on the Jetson.

Power Distribution Diagram:



8.2 Power Budget Estimation

Component	Voltage	Typical Current	Peak Current
Jetson Nano Orin	5V	1.5 A	3.0 A (AI inference burst)
12× MG996R Servos (walking)	6V	4.0 A (avg, 4 active)	12 A (all 12 stall — unlikely)
IMX219 Camera	3.3V (from Jetson)	0.15 A	0.2 A
PMS5003 Dust Sensor	5V	0.1 A	0.12 A

MQ-135 + ADS1115	5V	0.15 A	0.18 A
DHT22 + MPU6050 + GPS + HC-SR04	3.3–5V	0.15 A	0.2 A
TOTAL (Typical Walking)	—	~6.05 A	~15.7 A (worst case)

With a 2200 mAh battery at typical 6 A draw: Runtime ≈ (2.2 Ah / 6 A) × 60 ≈ 22 minutes. With a 3000 mAh battery: Runtime ≈ 30 minutes. These estimates are for continuous walking.

9. Wiring Guide & Pin Mapping (Jetson Nano Orin)

The Jetson Nano Orin Developer Kit has a 40-pin GPIO header compatible with the Raspberry Pi pinout. Below is the complete wiring map for every connected peripheral.

Peripheral	Signal	Jetson Pin	Wire Color (Suggested)	Notes
PCA9685 (I^2C)	SDA	Pin 3 (I2C1_SDA)	Yellow	I^2C Bus 1, 0x40 address
PCA9685 (I^2C)	SCL	Pin 5 (I2C1_SCL)	White	Pull-up resistors on PCA9685 board
PCA9685	GND	Pin 6 (GND)	Black	
PCA9685	VCC (Logic)	Pin 1 (3.3V)	Red	Logic power only; servo V+ from 6V buck
MPU6050 (I^2C)	SDA	Pin 3 (I2C1_SDA)	Yellow	Shares I^2C bus with PCA9685 (addr 0x68)
MPU6050 (I^2C)	SCL	Pin 5 (I2C1_SCL)	White	
ADS1115 (I^2C)	SDA	Pin 3 (I2C1_SDA)	Yellow	Shares I^2C bus (addr 0x48)
ADS1115 (I^2C)	SCL	Pin 5 (I2C1_SCL)	White	MQ-135 analog → ADS1115 → I^2C
DHT22	Data	Pin 7 (GPIO216)	Green	10kΩ pull-up to 3.3V
HC-SR04	Trig	Pin 11 (GPIO50)	Orange	
HC-SR04	Echo	Pin 13 (GPIO14)	Blue	Use voltage divider (5V→3.3V)
PMS5003	TX → Jetson RX	Pin 10 (UART1_RX)	Green	UART 9600 baud
NEO-6M GPS	TX → Jetson RX	Pin 8 (UART0_TX) — or USB	Green	Use USB-UART adapter if UART1 taken
IMX219 Camera	CSI Ribbon Cable	CSI Port (22-pin)	—	Official Jetson CSI connector
Power (5V Rail)	5V Input	Barrel Jack / USB-C	Red	From 5V/3A Buck Converter

IMPORTANT: The HC-SR04 Echo pin outputs 5V logic, but the Jetson GPIO is 3.3V tolerant only. Use a simple resistor voltage divider (1kΩ + 2kΩ) on the Echo line to step it down to 3.3V.

10. Sensor Integration & Placement

Correct sensor placement is critical for data quality. The following guidelines ensure each sensor operates in its optimal environment on the robot body.

Sensor	Mount Location	Orientation	Why This Position
IMX219 Camera	Front face plate, center	Lens facing forward, angled 15° downward	Captures leaf images at 0.3–0.6 m distance while walking
HC-SR04 Ultrasonic	Front face, beside camera	Facing forward, horizontal	Obstacle detection ahead of robot path
PMS5003 Dust Sensor	Top deck, sensor_top_mount.stl	Intake port facing forward (into airflow)	Clean airflow when robot moves forward; no recirculation
MQ-135 Gas Sensor	Top deck, sensor_top_mount.stl	Sensing element exposed upward	Exposed to ambient air; away from battery heat
DHT22 (Temp/Hum)	Top deck, shaded recess	Sensor element exposed, PCB shaded	Shaded from direct sunlight to prevent false high-temp readings
MPU6050 IMU	Center of body, base plate	Flat-mounted, aligned with body axes	Center of mass location for accurate balance measurement
NEO-6M GPS	Top deck, sensor_top_mount.stl	Ceramic antenna facing skyward	Unobstructed sky view for satellite signal reception

11. Mechanical Assembly Guide (Step-by-Step)

Follow this sequence to assemble the complete MVP quadruped robot:

Step 1 — Install Brass Heat-Set Inserts

Using a soldering iron tip (set to 220°C), press M3 brass heat-set inserts into all pre-designed holes in the body_base_plate, hip_bracket, and upper_leg parts. This creates durable, reusable threaded joints in the 3D prints. Install M2.5 inserts for Jetson mount holes.

Step 2 — Assemble Individual Legs (×4)

For each leg: (a) Mount the hip flex/ext servo into the hip_bracket using M3 bolts. (b) Attach the servo horn adapter to the MG996R horn. (c) Connect the upper_leg link to the hip servo horn. (d) Mount the knee servo into the upper_leg link. (e) Attach the lower_leg link to the knee servo horn. (f) Press-fit the TPU foot_pad onto the lower_leg tip.

Step 3 — Mount Hip Ab/Ad Servos in Body

Insert 4× hip abduction servos into the corner servo pockets on the body_base_plate. Secure with servo mounting screws. Route servo cables through internal wire channels.

Step 4 — Attach Legs to Body

Connect each assembled leg to its respective hip abduction servo horn. Use the servo_horn_adapter and M2 screws for a rigid connection. Verify smooth rotation range (±30°) on all 4 hip servos.

Step 5 — Install Electronics on Top Plate

Mount the Jetson Nano Orin on the body_top_plate using M2.5 standoffs (jetson_standoffs.stl). Mount the PCA9685 board on the pca9685_bracket. Mount the ADS1115 ADC with double-sided tape.

Step 6 — Wire Power Distribution

Solder the XT60 connector to the battery leads. Wire through the power switch. Split to two LM2596 buck converters. Set one to 5V (Jetson), one to 6V (servos). Connect 6V rail to PCA9685 V+. Connect 5V rail to Jetson barrel jack / USB-C power input.

Step 7 — Connect Sensor Wiring

Following the pin mapping in Section 9, connect all sensors to the Jetson GPIO header. Use dupont jumper wires. Install the voltage divider on the HC-SR04 Echo line. Route all cables neatly using wire_clip parts and zip ties.

Step 8 — Mount Sensors on Top Deck

Attach the sensor_top_mount plate. Insert PMS5003, MQ-135, DHT22, and GPS antenna into their designated slots. Ensure PMS5003 intake faces forward.

Step 9 — Install Camera & Ultrasonic

Mount the IMX219 on the camera_mount_gimbal, set tilt to ~15° down. Mount HC-SR04 on the body_front_plate. Connect the CSI ribbon cable to the Jetson's CSI port.

Step 10 — Close the Chassis & Install Battery

Slide the LiPo battery into the battery_tray under the base plate. Secure with velcro strap. Attach the body_top_plate to the body_base_plate using M3 screws. Snap on front and rear plates.

Step 11 — Servo Calibration

Power on with all servos connected. Run a calibration script that moves each servo to its center position (90°). Mark the neutral standing pose. Fine-tune offsets in the config file.

Step 12 — First Power-On & Smoke Test

Power on the robot. Verify: (a) Jetson boots to Ubuntu. (b) All 12 servos respond to I^2C commands via PCA9685. (c) Camera stream appears. (d) IMU reads accelerometer data. (e) All sensor nodes publish data. If everything checks out — the MVP is ready for gait testing!

12. Software Stack & ROS 2 Setup

12.1 OS & SDK Installation

4. **JetPack:** Flash NVIDIA JetPack 6.x SDK onto MicroSD card using NVIDIA SDK Manager.
5. **OS:** Ubuntu 20.04 LTS (included with JetPack).
6. **ROS 2:** Install ROS 2 Humble Hawksbill via Debian packages.
7. **AI Stack:** Install PyTorch for Jetson (aarch64), then ONNX and TensorRT Python bindings.

12.2 ROS 2 Node Architecture

The following ROS 2 nodes form the complete software pipeline:

Node Name	Function	Publishes Topic	Subscribes Topic
camera_node	Captures IMX219 frames	/camera/image_raw	—
disease_detector_node	Runs MobileNetV2 TensorRT inference	/disease/detection	/camera/image_raw
imu_node	Reads MPU6050 data	/imu/data	—
env_sensor_node	Reads DHT22, PMS5003, MQ-135	/environment/data	—
gps_node	Reads NEO-6M coordinates	/gps/fix	—
ultrasonic_node	Reads HC-SR04 distance	/ultrasonic/range	—
gait_controller_node	Executes trot/crawl gait patterns	/servo/commands	/imu/data
ik_engine_node	Converts foot positions → servo angles	/servo/angles	/gait/foot_targets
servo_driver_node	Sends PWM to PCA9685	—	/servo/angles
balance_pid_node	IMU-based body leveling	/balance/correction	/imu/data
telemetry_node	Publishes data via MQTT to dashboard	—	All sensor topics

12.3 Key Python Libraries

- **rclpy:** ROS 2 core client library.
- **OpenCV (cv2):** Image capture and preprocessing.
- **PyTorch:** Model training (desktop) → export ONNX.
- **tensorrt:** ONNX → TensorRT engine compilation and inference.
- **adafruit-circuitpython-pca9685:** PCA9685 servo driver communication.

- **smbus2:** I^2C communication (MPU6050, ADS1115).
- **paho-mqtt:** MQTT publish/subscribe for telemetry.
- **pyserial:** UART communication (PMS5003, GPS).
- **Adafruit_DHT:** DHT22 sensor library.
- **numpy:** Matrix math for IK and transforms.

13. Estimated Total Cost Breakdown

The following summarizes the total estimated cost for building one complete MVP quadruped prototype. Prices are approximate Indian market rates (2025).

Category	Components	Estimated Cost (₹)
Compute	Jetson Nano Orin + MicroSD + USB Wi-Fi	₹19,000 – ₹23,000
Sensors	Camera + Dust + Gas + Temp + IMU + GPS + Ultrasonic	₹3,000 – ₹3,800
Servos (MG996R ×12)	12 × MG996R Metal Gear Servos	₹3,000
Servos (DS3218 ×12 — alt)	12 × DS3218 High-Torque Servos	₹6,000
Power System	LiPo Battery + Charger + 2× Buck Conv + Switch + XT60	₹1,800 – ₹2,500
Electronics	PCA9685 + ADS1115 + Wires + Jumpers	₹600
3D Printing Filament	PLA+ (1 kg) + TPU (500 g)	₹1,400
Fasteners & Hardware	Bolts, nuts, inserts, velcro, zip ties, tape, heat shrink	₹760
TOTAL (with MG996R)		₹29,560 – ₹35,060
TOTAL (with DS3218)		₹32,560 – ₹38,060

Note: Costs are estimates and may vary. The Jetson Nano Orin is the single most expensive component. If your institution provides the Jetson board, the total drops to ₹11,000 – ₹16,000 for remaining hardware.

14. Open-Source References & CAD Resources

The following open-source quadruped robot projects provide excellent starting points for CAD files, kinematics code, and mechanical design inspiration. You can adapt their STL files to fit the MVP dimensions and sensor payload defined in this document.

Project	Description	URL / Repository	What to Use
SpotMicro	Most popular open-source 3D-printed quadruped. Full STL files, IK code, ROS integration.	github.com/michaelkubina/SpotMicroESP32 github.com/mike4192/spotMicro	Body shell, leg STLs, IK Python code. Resize to fit Jetson Nano Orin.
Mini Pupper (MangDang)	Commercial open-source quadruped kit. ROS 2 compatible. ~$300 USD.	github.com/mangdangroboticsclub/mini_pupper	ROS 2 launch files, gait controller, servo calibration scripts.
OpenQuadruped	Modular open-source quadruped for research. Python-based controller.	github.com/adham-elarabawy/open-quadruped	Gait patterns (crawl, trot), IK engine, simulation environment.
Stanford Doggo	Dynamic quadruped with quasi-direct-drive actuators.	github.com/Nate711/StanfordDoggoProject	Mechanical design principles, gait transition logic.
Champ (ROS)	ROS/ROS 2 quadruped locomotion controller framework.	github.com/chvmp/champ	Plug-and-play ROS 2 gait controller. Supports custom URDF robot models.
PlantVillage Dataset	The primary training dataset for crop disease detection AI.	kaggle.com/datasets/emmarex/plantdisease	Training images for MobileNetV2 disease detection model.
Fusion 360 (Free for Students)	Autodesk parametric CAD — free education license.	autodesk.com/products/fusion-360	Design custom body shell, modify leg dimensions, export STL.

This document provides a complete blueprint for building the MVP Precision Agriculture Quadruped Robot Dog from scratch. Follow the sections sequentially: print the parts, procure the BOM, assemble the hardware, wire the electronics, flash the software, and begin field testing. Good luck, Adithya & Rachit!